IOTAFLOW
SYSTEMS PVT. LTD

RPD OIL METER INSTALLATION MANUAL

" If You Want To Manage It, Measure It "

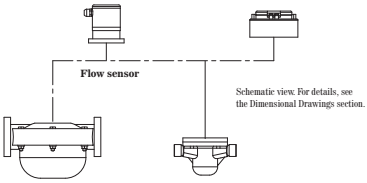
www.iotaflow.com

I. Summary

Metoil ®oil meter from **IOTAFLOW** has 2 parts mainly – hydraulic part (flow sensor) and measurement transducer with display, totalizer and signal output where applicable. The flow rate range determines the nominal size of the flow sensor. The measurement transducer is either electronic or mechanical.

Electronic Counter is a battery / 24 VDC operated, intelligent ultra-low power meter. The central processor adopts MSP430F147 chips, and piecewise compensation can be realized by setting the flowmeter parameters, which can improve the flow meter's range ratio greatly.

Mechanical counter transducer helps to realize totalized value only with a combination of mechanical gears to convert volumetric movement of piston to Ltrs / m3 as the case may be. A pulser can be added as and when required to convert totalized value in Pulse signal for remote reading or connect with BMS / SCADA etc.



The flow sensor and the measurement transducer are calibrated as single unit and form one metering unit. For optimal results in case of differential measurement, meters with special calibration should be used*. (* consult factory) If the measurement transducer is replaced at later stage, a certain divergence in measured values is to be expected.

Applications, Planning and installation

Metoil ®oil meters must be only be used for their intended purpose and comply with safety regulations. **IOTAFLOW** devices are manufactured according to valid standard and guidelines. **IOTAFLOW** guarantees the quality of the product in the context of its general terms of the business. The owner or operator will be liable for the correct installation as well as the appropriate handling of the equipment upon its receipt. The instructions for assembly and operation are to be followed exactly. None of the information stated here releases planner, installers and the operators from their own careful and comprehensive assessment of the respective plant configuration in terms of functional capability and operational safety.

Safety Precautions

- The design of the plant must ensure that the meter cannot be damaged, especially from the effects of icing, torsion in the piping, excessive heat expansion of the piping misaligned pipes during installation, instruction of foreign bodies or silt-up.
- The clearance distance of the piping must be adhered to when mounting the meter. If using flanged connections, the correct number of connector elements must be fitted and they must be tightened with the correct torque in accordance with the screw manufacturer's instruction.
- Comply with the permissible operating data as defined on the type plate. Pressure test with a maximum 1.5 x the nominal pressure (PN).
- Make sure that no hazardous fumes can build up in the piping and in the meter during commissioning, decommissioning and dismantling.
- The meter must at all time be completely filled with liquid during operation.
- Check meter periodically for tightness of the connections and for proper functioning.
- If work is to be done on the installation, before each intervention:
  1) Release the pressure in the installation.
  2) If hazardous liquids are used, wear protective clothing and safety goggles.
  3) Place a collecting tray underneath the installation.

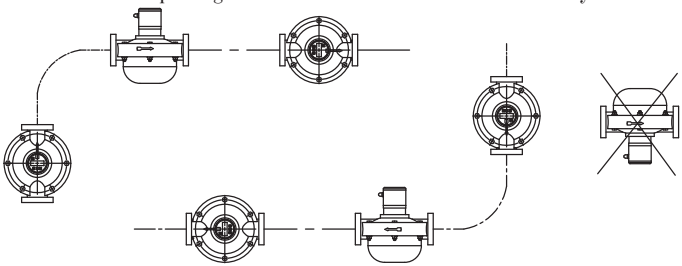
Plant design

Layout of piping

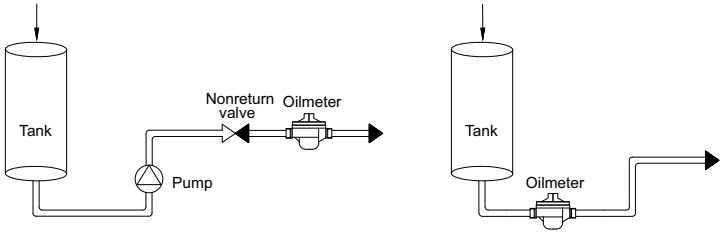
Meter installation

Easy access for reading the meter and controlling the ancillary equipment is important. Provide that the arrow on the housing is in the direction of flow. The meter can be installed in any position without any special modifications. The display is adjusted to the installed position in situ.

Do not mount the meter with the head pointing downwards. Flow conditioners are unnecessary.



The layout of piping must ensure that the meter is filled with liquid at all time and that no inclusions of air or gas may occur.



The quantities from all consumers must be registered by the meter.

Correct layout of meter and accessories.

If the meter is used for viscosities in excess of 5 mPa.s, or if it is mounted on the suction side of a pump, the pressure loss and the flow rate that can still be attained should be determined with the help of the pressure loss curves provided in Metoil technical information. In addition, the pressure loss due to installed filters must be taken into consideration.

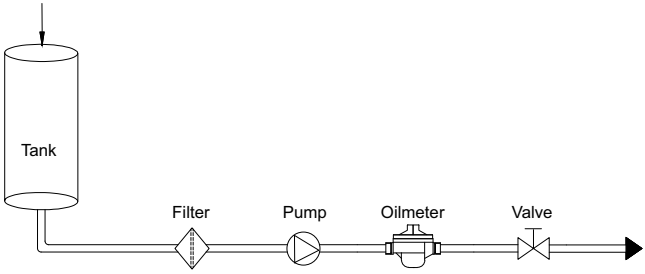
Select the meter and ancillaries according to the max, working conditions:

- Operating pressure and temperature.
- Ambient Temperature -10 degree Celsius…..60 degree Celsius
- Resistance of the material against liquid to be metered, working conditions.
- Flow rate.

The meters are to be selected according to the flow rate band not according to the pipe diameter.
If necessary, adapt the pipework.

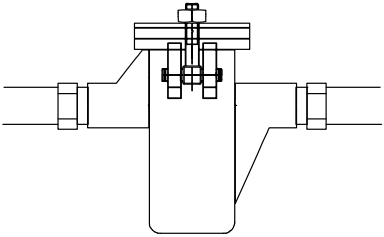
Stop valves or cocks

In order to avoid backflow and draining, stop values must be mounted after the meter. Backflow and draining cause faulty measurement and may damage the meter.



Dirt filter, Safety filter

Filters should be fitted to pervert any damage to the meter from impurities in the liquid.



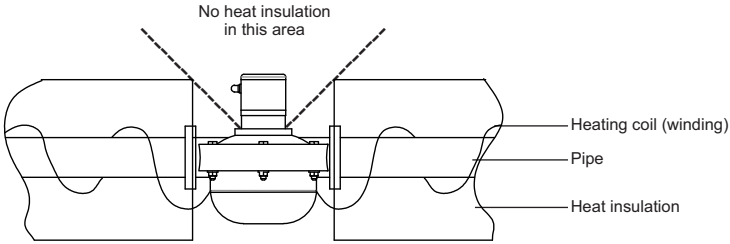
Maximum mesh width for filter		
Nominal Size	Meter type	
	RPD -D/M – 1% accurate	RPD -M – 0.5% accurate
DN15	0.250 mm	0.100 mm
DN20	0.400 mm	0.100 mm
DN25	0.400 mm	0.250 mm
DN40	0.600 mm	0.250 mm
DN50	0.600 mm	0.250 mm
DN 80	0.600 mm	0.250 mm

* If a dirt filter with the given mesh size is used, the safety filter in the meter inlet can be removed.

The Filter mounted in the meter inlet is only a safety filter and is too small to act as a dirt filter.

Heat insulation

The measurement transducer must not be insulated. This could cause its permitted temperature range to be exceeded



The permitted temperature ranges for the meter must be observed.

Special requirements – pulsers for remote evaluation

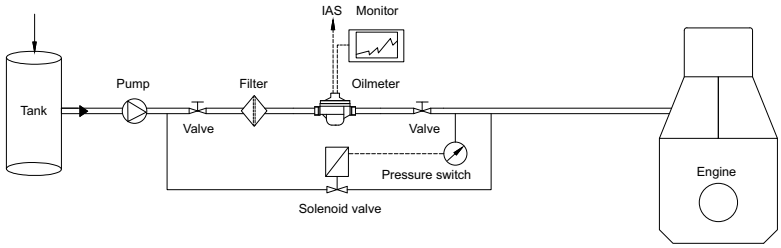
Return flow must be avoided for meter equipped with pulsers (RPD-M) / pulse out (RPD-D) for remote evaluation. If the plant design cannot ensure this, a non-return valve must be installed.

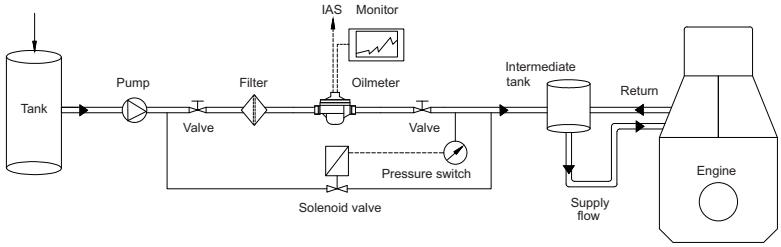
Special requirements – differential measurements

For differential measurements, one meter is installed in the pipe for supply and return flow respectively. The difference between the supply and return flow volumes determines the consumption.
If ordered with the 'differential measurement' option, RPD-D/M 0.5% accurate meters are calibrated in accordance with the indicated supply and return flow volumes. Make sure that these meters are installed in the correct pipes, i.e. the supply flow meter should be installed in the pipe that supplies the consumer and return flow meter should be installed in the return pipe.

Special requirements – ships

On ships, attention is needed to ensure that the engine can continue to operate at full power even if there is heavy filter contamination or is the meter is damaged. A pressure switch can be used to switch over to the bypass and to draw attention to the need for servicing. The engine then continues to operate but without consumption measurements.

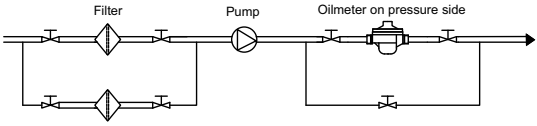


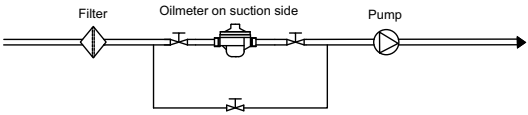


Ship classification societies require the installation of bypass pipes. The relevant must be followed.

Installation of the meter on the suction side of a pump

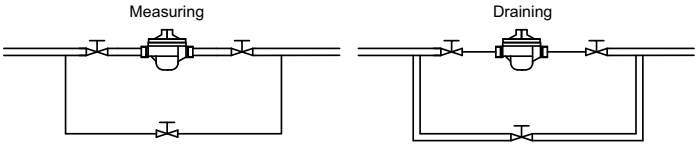
If the oil meter is installed on the suction side of the pump, consideration must be given to the maximum pressure loss at the maximum permitted flow rate and the maximum viscosity that occur. Installed filters must also be taken into account.





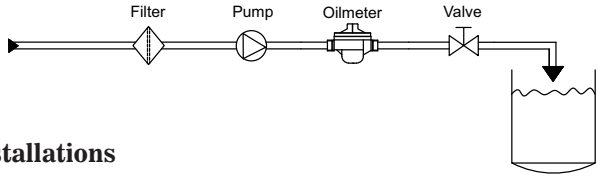
Draining pipes

If the pipe are to be drained with the compressed air at a later stage, stop calves must be provided on both sides of meter.



Special requirements – fillings and dosing units

For filling and dosing, the valve must be fitted between meter and discharge.
The shorter the pipe section between valve and discharge, the higher the accuracy. Fast opening and closing of the valves should be avoided as this can cause pressure
or water hammer. Water hammer damage the meter.



Electrical installations

Electrical installations are subject to legal regulations which must be observed when planning the facility. Electrical installations must only be carried out by authorized specialists.

In the layout of the installation, the following factors should be taken into accounts during plant design :

- Ancillary device connected to the meter
- Maximum cable lengths with/without repeater
- Junction boxes, cable guides
- Environmental disturbances

Electrical installations are subject to special regulations, which require mandatory compliance.

Installing the meters

Preparing for installation

Check meters and installation material.
Compare the data of the meter with the expected maximum conditions of the installation. They may not exceed the meter specifications:

- Maximum flow rate (Qmax l/h)
- Maximum operating pressure (PN bar)
- Maximum temperature (degree Celsius)
- Appropriate connections (threaded or flanged) and seals (gaskets)
- Fasteners for the meter
- Resistance to liquid to be metered and temperature

When existing plants are altered:
Flush the installation and put it out of operation. Close the stop valves of the section containing the meter.
Make sure that unauthorized start-up is not possible while mounting.

The applicable working regulations must be complied with the during all work on the plant.
Prepare the pipe and connections for the meter in the specified length for installation, and install intermediaries piece with a dirttrap where appropriate.

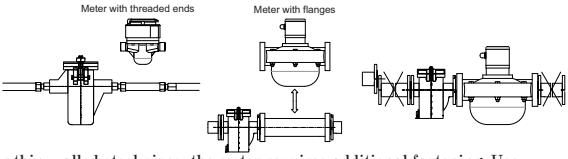
Start trial operation; open the stop valves **slowly** when doing this.

- Carry out a pressure test in the plant
- Check for leak tightness
- Flush the pipework until no residues are left
- Release the pressure and stop the plant again.

This trail operation ensures that the pipework is tight and clean.

Installation the meter in the pipe

Remove the protection plugs or caps from the meter (inlet and outlet).
Measurement of heavy fuel oil with RPD-D/M DN 20 or 25: If a dirt filter with a mesh sixe of max 0.5 mm is installed, the safety filter may be removed from the meter inlet to reduce the pressure loss.
Insert the meter into the pipeline in the prescribed position and flow direction. The arrow on the flow sensor should correspond with the direction of flow. Install mating flanges parallel and without tension in the pipe.



For pipes made of copper or thin-walled steel pipes, the meter requires additional fastening. Use appropriate fasteners.

If another pressure test is done after mounting the meter, the following pressure are acceptable for a short duration:

Nominal pr essure	Maximum Test pressure
16 bar	25 bar
25 bar	40 bar
40 bar	64 bar

Electrical connections

Safety precautions

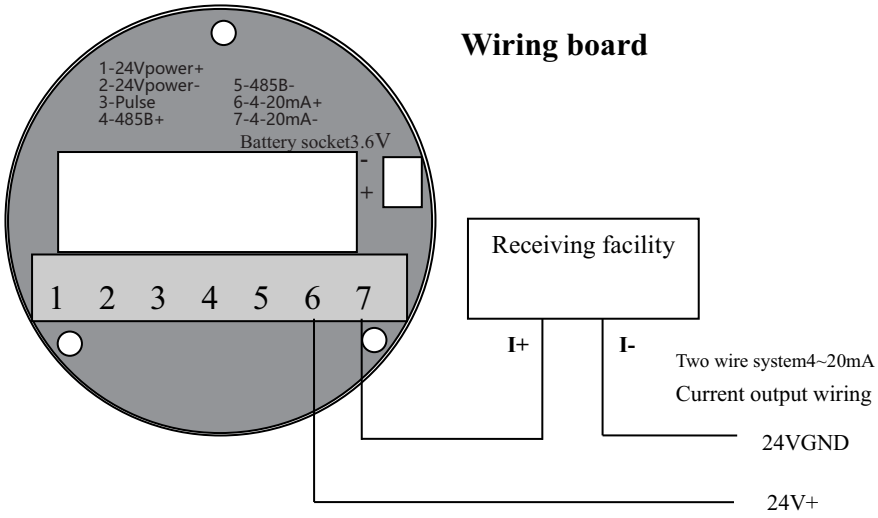
Disconnect from power supply, before working on electrical installations, make sure that nobody can power up the facility.
Pay attention to installation instructions for electrical devices:

- Voltage, operation data
- Maximum transmission length
- Cable cross section, length
- Ambient temperature, mounting position

Wiring diagram RPD-D and RPD-D 0.5% accurate

Our different output functions are available:

- Pulser for volume pulses with programmable pulse value (for external totalizer)
- Analogue current output 4….20 mA corresponding to flow rate

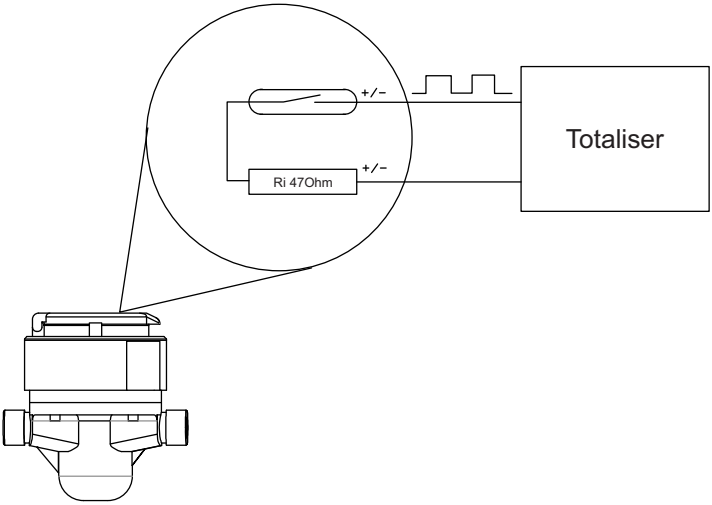


Options RPD-M and RPD-M

Reed type Pulser

Cable; 3 m, Polarity: arbitrary

Ambient temperature	-10 degree Celsius….70 degree Celsius
Switching element	reed contac t
Switching voltage	max. 48 V DC/AC
Switching current	max.50mA (Ri 47 Ω)
breaking capaci ty	max 2W
Quiescent current	zero
Pulse value	see dial plate



Parameterising ancillary devices

Some ancillary units require programming of pulse values (see the relevant operating instructions)
Pulse values of the meters: see dial plates.

Commissioning

Putting the plant into operation.

Open valves **slowly**, fill pipework gradually.
Vent the installation well.
Pressure shocks must be avoided in order not to damage the meter. Inclusions of air cause measuring errors in all types of meter and can damage them in operation.

Check the tightness of the connections of the device.

Check the flow rate of the installation.

Electronic display: read the instantaneous value.

Roller counter type meter: measurement of flow meter volume for 30….60 seconds

The flow rate is calculated using the following formula:

$$\frac{\text{Totalised valume in litres X 3600}}{\text{Measuring time in seconds}} = \text{Litres per hour}$$

Should the established flow rate be greater then the specification of the meter (Qmax), either a flow control valve (throttle) must be inserted behind the flow sensor or a large size flow sensor must be used.

Check functions of the connected accessories.

Servicing / repairs

Safety precautions

Before working on the hydraulics:

- Put the plant or section out of operation
- Close the stop valves
- Release the pressure

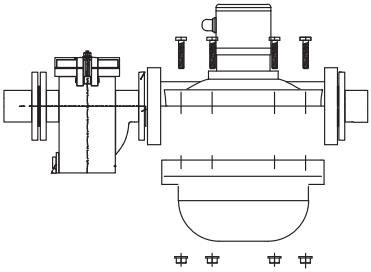
To restart the plan
- Slowly open the stop valves, avoiding pressure surges (‘water hammer’)
- Vent the pipe well
- Check tightness

Dirt filter

Dirt filter must be cleaned periodically, initially at short intervals.

Maintenance, spare parts

Check connections periodically for tightness and if necessary tighten again. For control and cleaning, the measuring chamber and the ring piston of the meter DN15….80 can be removed with dismantling the meter from the pipe.



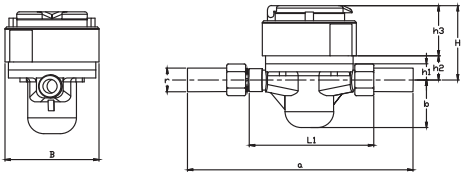
Torque of measuring chamber screws		
Flow meter	Screws	Torque
DN15.20	M 6	6 Nm
DN 25	M 8	16 Nm
DN 40	M 12	47 Nm
DN 80	M 16	100 Nm

The cleaning and revision cycle depends largely on the conditions of operation. Under favorable conditions 5….10 years suffice. Check the units for corrosion. When ancillary is used, observe their mounting and operation instructions. Spare part lists may be requested from **IOTAFLOW**.

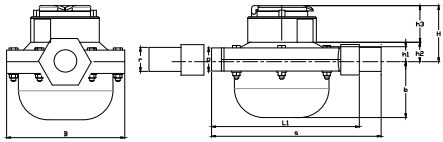
Dimensional drawings

Flow sensors

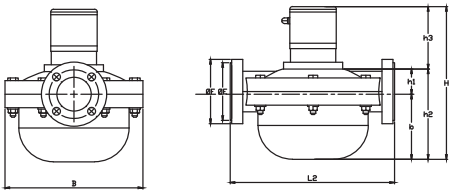
DN 15, 20, 25: with threaded ends



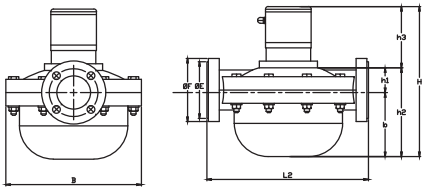
DN 40: with threaded ends



DN 15, 20, 25: with flanges (DN 2501/SN 21843)



DN 40, 50: with flanges (DN 2501/SN 21843)



Nominalsize	L1	L2	B	a	øE	øF	b	h1	h2	h3	H	p	r	Wight (kg)	Wight (kg)
RPD15	150	155	ø108	250	ø45	ø95	52.5	27	89.5	102	191.5	G$\frac{3}{4}$"	G$\frac{1}{2}$"	....	....
RPD20	165	170	ø108	265	ø58	ø105	53	27	90	102	192	G1"	G$\frac{3}{4}$"	3.500kg	....
RPD25	210	215	ø150	310	ø68	ø115	57	32	101	102	203	G1$\frac{1}{4}$"	G1"	7.500kg	....
RPD40	....	300	ø230	....	ø88	ø150	71.5	43	154.5	102	256.5	....	....	....	....
RPD50	....	320	ø245	....	ø102	ø165	78.5	99	178	102	280	....	....	....	....
RPD80	....	380	ø310	....	ø138	ø200	100	86	225.6	102	327.6	....	....	51.0kg	....

Dimensions in mm

Notes

IOTAFLOW
SYSTEMS PVT. LTD

RPD OIL METER INSTALLATION MANUAL

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